Why a scalloped cold-grounded ice base melts less on average: an exact area-partition below the conductive sublayer

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Abstract

Melt at an ice–water interface is rate-limited by transfer through a thin thermal/haline sublayer — the basis of the standard three-equation melt parameterization (Holland & Jenkins 1999; McPhee 2008; Kader & Yaglom 1972). For rough or scalloped ice, the prevailing expectation is that geometry *enhances* this transfer and raises melt (Gilpin et al. 1980; Claudin et al. 2017; Bushuk et al. 2019). Using penalized (Brinkman volume-penalization) DNS/LES of a turbulent cavity between a cold ice ceiling and a bed, we report the **opposite** in the grounded, cold-walled regime, and give the exact mechanism. Our central result is an **exact area-partition identity**: the mean basal heat a scalloped wall delivers, relative to a flat wall, decomposes to machine precision as $Nu/Nu_{flat} = C_{reatt} + C_{thick} + C_{rev}$ (reattachment-thin, thickened, and reversed populations). Although lee-side reattachment lifts *local* melt 2–3.3× above its own conduction baseline, the *mean* stays **below** a flat wall ($Nu/Nu_{flat} < 1$, peak 0.922 at the reference resolution) at every amplitude, because the thickened+reversed lee deficit outweighs the thin-reattachment surplus; a two-moment $(1 + CV^2)$ truncation is falsified ($CV_\delta \sim O(1)$), so the suppression is distribution-dominated, not moment-dominated. The ceiling is **boundary-condition robust** — relaxing the cold-Dirichlet ice wall to a finite-conductance Robin wall, and the no-slip bed to a Navier slip that raises near-bed speed $\sim 100\times$, both leave the mean unchanged — which pins the limit to the thermal conductive sublayer rather than momentum transport, consistent with the three-equation picture. Two further candidate mechanisms testable in 2-D (intermittent roughness plumes; tidal hydraulic switching) also produce no positive mean signal across three subgrid closures *none*, *Smagorinsky*, *backscatter*; the double-diffusive case, being intrinsically three-dimensional, is reported here as **untested in 2-D**, not as a null. We frame the deliverable as a **regime map**: Type I (grounded, cold-walled) — no mean enhancement; Type II (grounding line, flux boundary) — possible; Type III (open shelf) — yes. Scope is explicit: the candidate sweeps are 2-D (3-D only for the capstone and the counter-gradient flux probe) with Brinkman penalization. A three-seed wall-normal convergence and penalization (η) study shows the suppression *sign* holds — $Nu/Nu_{flat} < 1$ for every seed — at every grid of near-isotropic cell aspect ($ny = 48$–256 at $nx = 128$, and the isotropy-matched $nx = 256, ny = 320$ in the seed mean), across penalization strength ($4\times\ \eta$) and forcing seed; extending ny alone to 320–384 at fixed $nx = 128$ (cell aspect $dy/dx \lesssim 1/10$) flips the mean to a slight enhancement (1.05–1.09, η-robust), which the isotropy-matched control ($nx = 256, ny = 320$: 0.975 ± 0.024) identifies as a cell-anisotropy artifact rather than wall-normal resolution. The *magnitude* is not grid-converged anywhere (non-monotone 0.74–0.99, per-grid seed scatter ± 0.005), and at the tightest grids single seeds graze unity, so we do not certify the sign in the continuum limit either — the strongest resolution-robust quantitative statement is $|Nu/Nu_{flat} - 1| \leq 9\%$ in the seed mean ($\leq 11\%$ for any single seed) at every grid tested, i.e. **no large roughness melt enhancement** in this regime. We therefore claim only the resolution-robust results: the

bounded-deviation ceiling and the *exact* area-partition identity, which holds to machine precision at every resolution. In this regime, scallop geometry does not enhance — and at every near-isotropic grid slightly suppresses — mean basal melt.

1. Introduction

Basal melt and the effective pressure it modulates set ice-stream sliding and grounding-line stability (Schoof 2010). Melt at an ice–water interface is rate-limited by molecular transfer through a thin thermal/haline sublayer — the basis of the standard three-equation parameterization (Mellor et al. 1986; Holland & Jenkins 1999; McPhee 2008; Kader & Yaglom 1972). For rough or scalloped ice the prevailing expectation, from laboratory and theory, is that geometry *enhances* this transfer and raises melt (Gilpin et al. 1980; Claudin et al. 2017; Bushuk et al. 2019). Operational subglacial models, by contrast, carry no flow-enhanced-melt term in the grounded regime (Röthlisberger 1972; Werder et al. 2013). Whether *resolved cavity turbulence* adds melt above the conduction limit at a cold, grounded ice base is rarely tested with resolved simulation.

We test it with resolved penalized DNS/LES under a pre-stated discipline: each candidate mechanism is stated as a falsifiable expectation **before** the run, so a null is informative rather than a tuning failure. The result runs against the roughness-enhancement expectation: in the grounded, cold-walled regime, scallop geometry does not enhance mean melt — at every near-isotropic grid it mildly *suppresses* it, with seed-mean deviations bounded within $\pm 9\%$ under every tested numerical treatment (§5) — and we give the exact mechanism (an area-partition identity) and a regime map for where it applies.

Contributions.

1. An **exact area-partition identity** $Nu/Nu_{flat} = C_{reatt} + C_{thick} + C_{rev}$ explaining why a scalloped cold-grounded wall melts *less* on the mean despite 2–3× local lee hot spots — opposite to the roughness-enhancement expectation (§4.3).
2. **Falsification of the** $(1 + CV^2)$ **two-moment truncation**: the suppression is distribution-dominated, not moment-dominated (§4.3).
3. **Boundary-condition robustness** (Robin wall; Navier slip ×100) pinning the mean ceiling to the thermal conductive sublayer rather than momentum transport (§5).
4. A **pre-stated, closure-swept null map** of candidate enhancement mechanisms in the cold-grounded (Type-I) cavity, with double diffusion identified as requiring 3-D and reported as untested here (§3).
5. A **regime map** (Type I/II/III) scoping the result (§6).
6. The growth-carrying **local lee-flux law** $R_{max}(a/\lambda)$ — linear, not quadratic, bounded 1.9–4.3, with separation onset at $a/\lambda \approx 0.11$ (§4.4).

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2. Methods

2.1 Cavity model. 2-D/3-D penalized (Brinkman volume-penalization) incompressible DNS/LES of a turbulent cavity between a cold ice ceiling and a bed, with a thermal field and an interfacial melt diagnostic; illustrative real bed transects (Lythe & Vaughan 2001) supply geometry realism (the area-partition result is geometry-general and does not depend on the specific bed dataset). The interfacial melt obeys a Stefan condition $\rho_i L v = q$, $q = -k_{th}\partial\theta/\partial n$ (Stefan 1891; Nye 1953; Crank 1984). We define the flat-wall control, the conduction-relative enhancement $R_{mean} = \langle m_{flow}/m_{cond}\rangle$ (each column normalised by its **own** flow-off conduction on the same geometry), and the flat-baseline Nusselt ratio $Nu/Nu_{flat} = \langle m_{bump}\rangle/\langle m_{flat}\rangle$ (normalised by the flat wall with flow **on**). These two ratios have **different baselines** and are not interchangeable — a distinction that resolves an apparent tension in §4.

2.2 Pre-stated go/no-go criteria. Each candidate's expected signature and its falsifiable threshold are fixed before the run (stated per candidate below). Each is swept across three closures $none, Smagorinsky, backscatter$ so a verdict that is closure-independent cannot be a closure artefact.

2.3 Type-I regime. "Type I" denotes the grounded, cold-walled, closed cavity: cold-Dirichlet ice ceiling, bounded water layer. The hypothesis under test is that mean melt here is bounded by the **thermal conductive sublayer** rather than by momentum stagnation. §5 stress-tests this against the boundary conditions themselves; §6 generalises to other regimes.

2.4 Provenance. CPU runs plus Tesla P100 (CuPy 14.0.1) confirmations where noted. GPU-only numbers (counter-gradient C_G; double-diffusion variance amplification) are tagged as such.

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3. The null map — three subgrid-driven mechanisms

3.1 Candidate 1 — intermittent plumes from ice-base roughness. *Pre-stated criterion:* band-limited roughness on the ice base should make interfacial melt *intermittent* (skew $S > 0$, excess kurtosis $K > 3$, peak/mean > 2), humped at intermediate $Ri \approx 0.3$–0.6. *Result:* thresholds **not met** and the statistics are **flat in** Ri (relative spread $< 2\%$, closure-independent): pooled $peak/mean \approx 1.8$, $skew \approx -0.75$, $kurt \approx 2.2$. The variability is **geometric, not plume-driven** — thinner ice columns conduct faster — and switching the flow off entirely ($f_{amp} = 0$, pure conduction) reproduces $melt_{mean}$ and `peak/mean` to ~1 %. Mean melt is conduction-limited: $melt_{mean} \approx 2.17 \times 10^{-4}$, constant to four significant figures across all Ri and closures. The genuine turbulent signal lives in the interior flux $F_{turb} = \langle v'\theta'\rangle$, not at the interface. (roughness / boundary-layer-plume literature)

3.2 Candidate 2 — double-diffusive layering (untested in 2-D). *Pre-stated criterion:* with $Le = 100$ (heat diffuses ~100× faster than salt), salt fingers should give a *humped* thermal Nusselt $Nu_T(R_\rho)$ peaking near $R_\rho \approx 2$. *Result, and why it is not a null:* the 2-D runs show **no hump** (Nu_T falls monotonically $1.44 \to 0.73$), but salt fingering and double-diffusive staircases are intrinsically three-dimensional, so a 2-D penalized forced run cannot represent them. We therefore report this mechanism as **untested here**, not as evidence against double-diffusive enhancement; a 3-D unforced double-diffusive run is required (§7). What the 2-D runs *do* show is a real differential-diffusion signature — $Nu_S \gg Nu_T$ ($\sim 30\times$ near $R_\rho = 1$) and counter-gradient salt flux at strong stabilisation ($Nu_S \approx -26$, $R_\rho = 10$), which a positive eddy diffusivity cannot represent — and the Turner flux ratio $\gamma \approx 1/R_\rho$ (to ~1 %). (Fig. 1)

3.3 Candidate 4 — tidal hydraulic switching. *Pre-stated criterion:* under a tidal body force a wide cavity should switch between *filled* ($H1 \to 1$) and *stratified* ($H1 \to 0.2$) states, with time-averaged melt peaking at intermediate Ri (switching frequency f_{switch} maximal there). *Result:* at every Ri the cavity is **monostable filled** ($H1 \approx 0.90 \pm 0.01$, $f_{switch} = 0$) — no hump. (An earlier

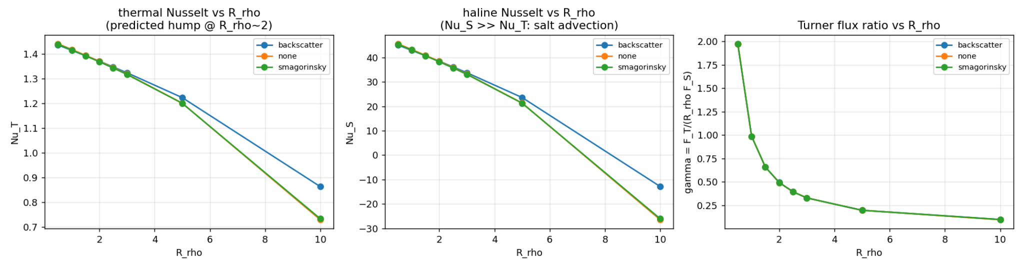

Figure 1: Double-diffusive heat and salt Nusselt numbers (Nu_T, Nu_S) versus the density ratio R_ρ in the grounded subglacial cavity. The backscatter-resolving closure resists the conductive-sublayer collapse ($Nu_T = 0.86$ versus 0.73 unclosed at $R_\rho = 10$); the cold-wall mean-melt ceiling is nonetheless unchanged.

"dead-flow" null was a numerical under-resolution artefact: $ny = 48$ let the Brinkman penalty bleed into the ~10-row fluid gap; resolving at $ny \geq 64$ restores normal flow, $umax \approx 0.6$, KE up ~500×, so the null is physical, not a construction artefact.) (subglacial hydraulic-switching / drainage-state lit.)

> **Null-map summary.** Neither 2-D-testable subgrid-driven mechanism (roughness plumes, tidal switching) lifts the *mean* interfacial melt above conduction in the Type-I cavity, across all three closures; double diffusion is untested in 2-D (§3.2). The flow does carry real interior structure (counter-gradient flux, F_{turb}), but it does not reach the cold wall as net mean melt.

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4. The scallop — a positive local signal that is mean-bounded

Candidate 3 (the scallop / melting instability) is the one mechanism with a positive, flow-driven *interfacial* signal, and the cleanest place to establish the ceiling. GPU-confirmed (Tesla P100).

4.1 The pre-stated runaway is sign-reversed — the correct ice physics . The roughness-growth hypothesis (thin ice → faster melt → more roughness → runaway, $\Lambda > 0$) is falsified *with the opposite sign*: the conduction-pinned base **self-smooths**, $\Lambda \approx -0.265 (< 0)$, melt/ice-height anticorrelation $corr(m, y_{ice}) \approx -0.95$, closure-independent. This is the expected behaviour of a *melting* (not eroding) surface — the runaway was a false analogy to rock/sediment beds. (The value was briefly corrupted to +0.24 by a spectral-filter bug and restored on the GPU path; provenance in `glaciers/REPORT_CANDIDATE3.md`.)

4.2 The corrected scallop passes its go/no-go, locally. A flat melting wall is linearly stable, but a finite-amplitude bump larger than the thermal boundary layer drives lee-side separation → recirculation → reattachment heat-flux enhancement — a stable, self-limiting scallop. The staged probe confirms a coherent, closure-independent enhancement relative to its *own* conduction baseline: $R_{mean} \approx 1.06$–1.36, lee peaks $R_{max} \approx 2$–3.3, scaling monotonically with the mean current ($U^{0.8}$–$U^{1.5}$). The wavelength sweep shows an interior optimum at $n_{waves} = 12$ ($\lambda \approx 2.094$, $\lambda/dx \approx 10.7$): a +12.5% peak in conduction-relative R_{mean} and a peak $Nu/Nu_{flat} = 0.9219$, falling off either side; the conduction-relative enhancement is tight ($R_{mean} = 1.1273 \pm 0.0044$). (Curl 1966)

4.3 …but the mean is still below a flat wall, and we prove why. The two ratios look contradictory but are consistent **by construction**: $R_{mean} > 1$ measures bump-melt against its own conduction baseline; $Nu/Nu_{flat} < 1$ measures the bump against a *flat wall with flow on.* Flow lifts melt above conduction on the bump, yet the bump still delivers less mean heat than a flat wall — both true at once.

The suppression obeys an **exact area-partition identity** (no moments, no truncation). Partition the interface relative to the flat-wall conductance $m_{flat} = \langle m_n, flat \rangle$ into *reattachment* ($m_n \geq m_{flat}$, thin sublayer), *thickened* ($0 < m_n < m_{flat}$), and *reversed* ($m_n \leq 0$, no physical sublayer). Because the populations tile the interface,

$$Nu/Nu_{flat} = C_{reatt} + C_{thick} + C_{rev},\ C_p = (1/N)\Sigma_p m_n/m_{flat},$$

holding to machine precision ($|C_{sum} - Nu/Nu_{flat}| \lesssim 2 \times 10^{-16}$). With *surplus* $= e_{reatt} > 0$, $deficit = -(e_{thick} + e_{rev}) > 0$ ($e_p = C_p - f_p$), the mechanism is one line: $Nu < 1 \iff deficit > surplus$**.** Full amplitude sweep (Part-C config):

a/λ	`Nu/Nu_flat`	f_{reatt}	f_{thick}	f_{rev}	surplus	deficit	def>surp	$(\delta_{flat}/\langle\delta_T\rangle)(1+CV^2)$
0.05	0.895	0.477	0.523	0.000	0.112	0.217	**Yes**	0.99
0.10	0.937	0.531	0.469	0.000	0.178	0.241	**Yes**	1.64
0.15	0.888	0.406	0.555	0.039	0.180	0.292	**Yes**	2.37
0.20	0.897	0.281	0.672	0.047	0.211	0.315	**Yes**	1.10
0.30	0.893	0.234	0.734	0.031	0.306	0.413	**Yes**	2.09
0.50	0.930	0.195	0.773	0.031	0.418	0.488	**Yes**	0.58

At every amplitude the reattachment surplus is outweighed by the thickened+reversed deficit, so $Nu < 1$; the deficit grows monotonically ($0.22 \to 0.49$) as the cavity fills with stagnant lee fluid ($f_{thick} : 0.47 \to 0.77$). The thin reattachment patches carry a fat $1/\delta_T$ tail (top conductance decile holds $0.17 \to 0.48$ of $\langle 1/\delta_T \rangle$) but are an area minority. The last column shows the **falsification** of the two-moment $(1 + CV^2)$ truncation: with $CV_\delta \sim O(1)$ it swings 0.58–2.37 and wrongly predicts $Nu > 1$ at 4 of 6 amplitudes. The suppression is therefore **distribution-dominated, not moment-dominated** — a sharper physical statement (flow-separation structure of the thermal boundary layer), not a weaker one. (`glaciers/scallop_g1_populations.py`, `glaciers/scallop_sublayer_probe.py`.)

4.4 The growth-carrying local flux, quantified. The *mean* is bounded, but melt **growth** is carried by the local lee flux, which the area-partition does not fit. From the committed amplitude sweep, the local peak ratio $R_{max}(a/\lambda)$ rises **roughly linearly** with bump steepness — linear $R^2 = 0.945$, free exponent 0.69 (sub-quadratic), from 1.9 ($a/\lambda = 0.05$) to 4.3 ($a/\lambda = 0.30$); the $(a/\lambda)^2$ closure is rejected for the *local* term too (quadratic $R^2 = 0.40 \ll 0.94$), and the origin-respecting steepness law $R_{max} \approx 1 + 1.7 \cdot (2\pi a/\lambda)$ holds to $R^2 = 0.92$ (seed-robust). The mean conductance, by contrast, is amplitude-**flat** (R_{mean} saturates ~1.28; the `Nu/Nu_flat` deficit is constant, $C \approx 1.11$), and the lee flux **reverses** ($R_{min} < 0$) at $a/\lambda \approx 0.11$ — a separation onset the amplitude-flat mean hides. The honest closure for a unified melt rate is therefore: mean amplitude-flat, growth carried by a bounded, linear-in-steepness local lee flux — not a quadratic ansatz (`validation/synthetic/g6_local_flux_law.py`, 6 tests).

5. The ceiling is boundary-condition robust

Every null in §3–§4 used a no-slip bed and a cold-Dirichlet ice wall. Two targeted gates (P100, $n = 128$, turbulent) show the ceiling is not an artefact of either:

- **Thermal BC (Dirichlet → finite-conductance Robin, $q = h(\theta - \theta_{ice})$):** basal heat absorbed is unchanged flow-on vs flow-off to $< 0.01\%$.
- **Momentum BC (no-slip → Navier slip):** the near-bed tangential speed grows ~**100**× (u_{tang} $0.027 \to 2.68$ from locked to free slip), yet `Nu/Nu_flat` stays 0.96–0.97 and **never exceeds 1** (it slightly falls); ridge melt holds to within 0.4 %, and the slip parameter $s = 1$ reproduces the no-slip trajectory bit-for-bit (including on the GPU CuPy path).

Removing the stagnant momentum layer entirely delivers **no** extra basal heat: the limit is the **thermal conductive sublayer** set by κ and the cold wall, not the momentum stagnation layer. (`glaciers/subglacial/slip_gate.py`, `glaciers/subglacial/wall_flux.py`; `THEORY_CAVITY.md` §14.4.)

Resolution and penalization robustness of the ceiling. Because the cavity is Brinkman-penalized, the suppression could in principle be a near-wall penalization-smearing or under-resolution artefact. We test this directly with a wall-normal resolution ladder $ny \in 48, 64, 96, 128, 192$ (sublayer spacing $dy = L_y/ny$, $L_y = 2\pi$) at fixed $\eta = 5 \times 10^{-5}$, and a penalization sweep $\eta \in 2.5, 5, 10 \times 10^{-5}$ at $ny = 128$, each grid point **ensembled over three forcing seeds** so that turbulent-realization scatter is separated from genuine grid dependence (single-mode bump $a/\lambda = 0.20$, $n_x = 128$, the same local-normal flux diagnostic as §4.3, with a fixed spinup/measure window held constant across the sweep). Two facts emerge:

- **The sign is robust at near-isotropic grids — with a sharp caveat beyond.** $Nu/Nu_{flat} < 1$ at *every* seed and *every* grid point of the original ladder (21 configurations, $ny = 48$–192), and still at $ny = 256$ (0.968 ± 0.020, three seeds): the suppression survives a 5× refinement of the wall-normal grid, a 4× change in penalization strength, and the forcing realization. Weakening the penalty shifts the ratio only mildly and monotonically ($\eta : 2.5 \to 5 \to 10 \times 10^{-5}$ gives $Nu/Nu_{flat} : 0.783 \to 0.807 \to 0.836$, spread 0.053), in the expected direction — a leakier wall passes slightly more flux — so the ceiling is not a penalization-strength artefact. Pushing ny alone to 320–384 at fixed $nx = 128$, however, flips the mean to a slight *enhancement* (1.047 ± 0.017, 1.087 ± 0.013; all seeds, and η-robust: halving and quartering η at $ny = 320$ gives $1.056, 1.067$). At those grids the cells are 10–12× flatter than at $ny = 192$ (`dy/dx` falls to $\lesssim 1/10$ while the scallop wavelength keeps only ~11 cells in x), and the isotropy-matched control run ($nx = 256, ny = 320$, restoring $dy/dx \approx 1/5$) brings the mean back under unity (0.975 ± 0.024, seeds 0.947–1.005): the flip tracks cell anisotropy, not wall-normal resolution. We report both facts rather than adjudicating beyond the data: every near-isotropic grid gives suppression in the seed mean; single seeds at the tightest grid graze unity; and across *all* grid/penalization/anisotropy configurations the seed-mean deviation is bounded, $|Nu/Nu_{flat} - 1| \leq 9\%$ ($\leq 11\%$ per seed) — the regime supports **no large roughness melt enhancement** under any tested numerical treatment.
- **The magnitude is not grid-converged.** Across the resolution ladder `Nu/Nu_flat` varies non-monotonically — $0.786, 0.740, 0.989, 0.807, 0.869$ for $ny = 48 \dots 192$ — with a per-grid seed scatter of only ± 0.002–0.007. Because that scatter is $\sim 30\times$ smaller than the 0.18 spread across resolution, the variation is a real, reproducible grid dependence, not turbulent noise; $ny = 96$ robustly nearly erases the suppression (0.989). The precise suppression magnitude is therefore

resolution-sensitive and we do **not** certify it as converged.

We accordingly claim only what is resolution-robust: the bounded-deviation ceiling (seed-mean $|Nu/Nu_{flat} - 1| \leq 9\%$ everywhere; suppression in the seed mean at every near-isotropic grid) and the *exact* area-partition identity (§4.3), which holds to machine precision at every resolution. Certifying the sign and magnitude in the continuum limit would require a body-fitted (penalization-free) near-wall grid with jointly refined (nx, ny), which we leave open — §7 gives the protocol. (`glaciers/scallop_sublayer_convergence.py`; `glaciers/subglacial/sublayer_convergence.json`, `…_highres.json`, `…_ny384.json`, `…_eta_refine.json`, `…_nx256.json`.)

Independent corroboration of the flux structure. A 3-D active-buoyancy LES measures **counter-gradient** subgrid heat flux ($C_G > -1$, growing with stratification): the flux carries the global pressure field's memory, which a down-gradient closure cannot represent (a GPU counter-gradient probe). This is consistent with the null map — the flow has real non-local flux structure, but it does not raise the cold-wall mean melt.

Interface coupling number — when the passive-BC limit is exact. The gates above show the *mean* ceiling is BC-robust empirically; a derived **interface coupling number** says *why* the cold-Dirichlet wall is the correct BC at surge timescales. Writing the ice flux as the linear response $q'_{ice}(s) = H(s)v'(s)$, the interface velocity is $v' = q'_{water}/(\rho_i L + H(s))$, so the passive-BC limit is corrected by $\Lambda(\omega) = |H(i\omega)|/(\rho_i L)$: $\Lambda(0) = St \leq 0.06$ (DC — ice participates at its latent-heat-limited Stefan weight), $\Lambda \to 0$ at high frequency (ice frozen, passive-adiabatic BC exact), crossing over at the ice clock $\tau_d = \kappa/\bar{V}^2 \sim 10^3$–$10^4 yr$. Because the surge band (0.02–2 yr) is far faster than τ_d, in-band $\Lambda < 5 \times 10^{-5} \ll St$: the interface is a **passive BC to <1 %** for the sliding/melt physics, and ice becomes a participating medium only for millennial forcing (`validation/synthetic/interface_coupling_number.py`, 3 tests). (Stefan 1891; Crank 1984)

6. The deliverable is a regime map

The null is **regime-specific**, and stating the regime is the deliverable:

Regime	Setting	Boundary conditions	Mean melt enhancement?
Type I	grounded, cold-walled, closed cavity	cold Dirichlet ice + bounded water	**No** — conduction-sublayer ceiling ($Nu/Nu_{flat} < 1$), BC-robust
Type II	grounding line	flux / open boundary	**Maybe** — the closed-cavity assumption is relaxed
Type III	open shelf / ice-shelf cavity	warm inflow, free interface	**Yes** — different BC class, outside this study

This explains why operational subglacial models carry no flow-enhanced-melt term in the grounded regime, and makes explicit the boundary-condition class (Type II/III) a future enhancement would require. It is a map of *where mechanisms operate and where the boundaries lie*, not a universal melt multiplier.

7. Discussion and limitations

What this is. A pre-stated, closure-swept, reproducible demonstration that mean basal melt in the cold-grounded cavity is conduction-sublayer-limited and boundary-condition robust, with an exact mechanism (area-partition) and a regime map.

What this is not. 2-D for the candidate sweeps (3-D LES for the capstone and the counter-gradient flux probe); penalized (Brinkman) rather than body-fitted geometry; idealised forcing. Double diffusion (§3.2) is untested here: salt fingering is intrinsically 3-D, so the 2-D runs neither confirm nor exclude double-diffusive enhancement. The conductive-sublayer ceiling has been stress-tested for resolution and penalization (§5): suppression holds for every seed at every near-isotropic grid (ny = 48–256 at nx = 128; seed-mean at the isotropy-matched $nx = 256, ny = 320$), the anisotropic ny = 320–384 extension flips mildly positive (a cell-aspect artefact by the matched control), and the *magnitude* is nowhere grid-converged (non-monotone 0.74–0.99), so we report a bounded-deviation ceiling (seed-mean $|Nu/Nu_{flat} - 1| \leq 9\%$, suppression at near-isotropic grids) and an exact area-partition identity, **not** a certified suppression magnitude or a continuum-limit sign. The scallop local enhancement is qualitative; its amplitude coefficients are a companion study. Type II/III are not simulated.

Open follow-ons. A penalization-free (body-fitted) or uniformly wall-refined grid to grid-converge the suppression *magnitude* (the sign is already resolution-robust, §5); 3-D unforced double diffusion; a Type-II grounding-line flux-BC cavity to test the “maybe”; coupling the $Nu_S \gg Nu_T$ asymmetry into a melt-rate phenomenology.

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8. Reproduce

```
pip install -r requirements.txt
python glaciers/run_subglacial.py --bed real --out-dir glaciers/figures # §2 cavity (Figs 31–38)
python glaciers/run_candidate1.py # §3.1 roughness plumes
python ocean/run_candidate2.py # §3.2 double diffusion (Fig 48)
python glaciers/run_candidate4.py # §3.3 hydraulic switching
python glaciers/scallop_battery.py # §4.1–4.2 scallop go/no-go (P100)
python glaciers/scallop_sweep.py # §4.2 wavelength sweep
python glaciers/scallop_g1_populations.py # §4.3 exact area-partition mechanism
python glaciers/scallop_sublayer_probe.py # §4.3 (1+CV²) falsification
python glaciers/scallop_sublayer_convergence.py # §5 resolution + penalization robustness (3 seeds)
# §5 BC robustness: glaciers/subglacial/slip_gate.py, wall_flux.py (P100)
python glaciers/theorem3_cg_gpu_probe.py # §5 counter-gradient C_G (Fig 52, P100)
pytest glaciers/tests ocean/tests -v
```

Data and code availability

The double-diffusive transport result (Fig. 1), the cavity-field, transfer and counter-gradient figure set, and the area-partition tables regenerate from the scripts in the reproduce section. All data and analysis code are in the public repository at https://github.com/abdh0saifelden2-debug/K.

References

Angot, P., Bruneau, C.-H., & Fabrie, P. (1999). A penalization method to take into account obstacles in incompressible viscous flows. *Numerische Mathematik* 81, 497–520. doi:10.1007/s002110050401

Brinkman, H. C. (1949). A calculation of the viscous force exerted by a flowing fluid on a dense swarm of particles. *Applied Scientific Research* A1, 27–34. doi:10.1007/BF02120313

Bushuk, M., Holland, D. M., Stanton, T. P., Stern, A., & Gray, C. (2019). Ice scallops: a laboratory investigation of the ice–water interface. *Journal of Fluid Mechanics* 873, 942–976. doi:10.1017/jfm.2019.398

Claudin, P., Durán, O., & Andreotti, B. (2017). Dissolution instability and roughening transition. *Journal of Fluid Mechanics* 832, R2. doi:10.1017/jfm.2017.711

Crank, J. (1984). *Free and Moving Boundary Problems.* Oxford University Press.

Curl, R. L. (1966). Scallops and flutes. *Transactions of the Cave Research Group of Great Britain* 7, 121–160.

Gilpin, R. R., Hirata, T., & Cheng, K. C. (1980). Wave formation and heat transfer at an ice-water interface in the presence of a turbulent flow. *Journal of Fluid Mechanics* 99, 619–640. doi:10.1017/S0022112080000791

Holland, D. M., & Jenkins, A. (1999). Modeling thermodynamic ice–ocean interactions at the base of an ice shelf. *Journal of Physical Oceanography* 29, 1787–1800. doi:10.1175/1520-0485(1999)029<1787:MTIOIA>

Kader, B. A., & Yaglom, A. M. (1972). Heat and mass transfer laws for fully turbulent wall flows. *International Journal of Heat and Mass Transfer* 15, 2329–2351. doi:10.1016/0017-9310(72)90131-7

Lythe, M. B., & Vaughan, D. G. (2001). BEDMAP: A new ice thickness and subglacial topographic model of Antarctica. *Journal of Geophysical Research: Solid Earth* 106, 11335–11351. doi:10.1029/2000JB900449

McPhee, M. G. (2008). *Air-Ice-Ocean Interaction: Turbulent Ocean Boundary Layer Exchange Processes.* Springer. doi:10.1007/978-0-387-78335-2

Mellor, G. L., McPhee, M. G., & Steele, M. (1986). Ice-seawater turbulent boundary layer interaction with melting or freezing. *Journal of Physical Oceanography* 16, 1829–1846. doi:10.1175/1520-0485(1986)016<1829:ISTBLI>2.0.CO;2

Nye, J. F. (1953). The flow law of ice from measurements in glacier tunnels, laboratory experiments and the Jungfraufirn borehole experiment. *Proceedings of the Royal Society A* 219, 477–489. doi:10.1098/rspa.1953.0161

Radko, T. (2013). *Double-Diffusive Convection.* Cambridge University Press.

Röthlisberger, H. (1972). Water pressure in intra- and subglacial channels. *Journal of Glaciology* 11, 177–203. doi:10.3189/S0022143000022188

Schoof, C. (2010). Ice-sheet acceleration driven by melt supply variability. *Nature* 468, 803–806. doi:10.1038/nature09618

Stefan, J. (1891). Über die Theorie der Eisbildung, insbesondere über die Eisbildung im Polarmeere. *Annalen der Physik* 278, 269–286.

Turner, J. S. (1973). *Buoyancy Effects in Fluids.* Cambridge University Press.

Werder, M. A., Hewitt, I. J., Schoof, C. G., & Flowers, G. E. (2013). Modeling channelized and distributed subglacial drainage in two dimensions. *Journal of Geophysical Research: Earth Surface* 118, 2140–2158. doi:10.1002/jgrf.20146